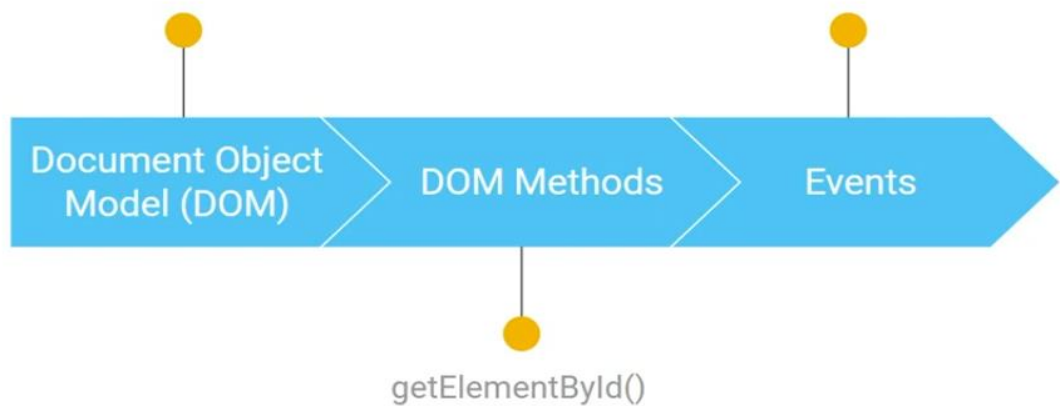


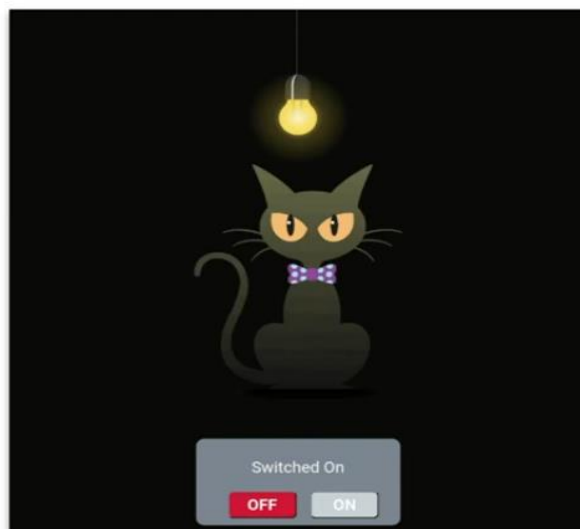
DOM and Event Fundamentals

Manipulating HTML and CSS
HTML DOM Tree

onclick Event

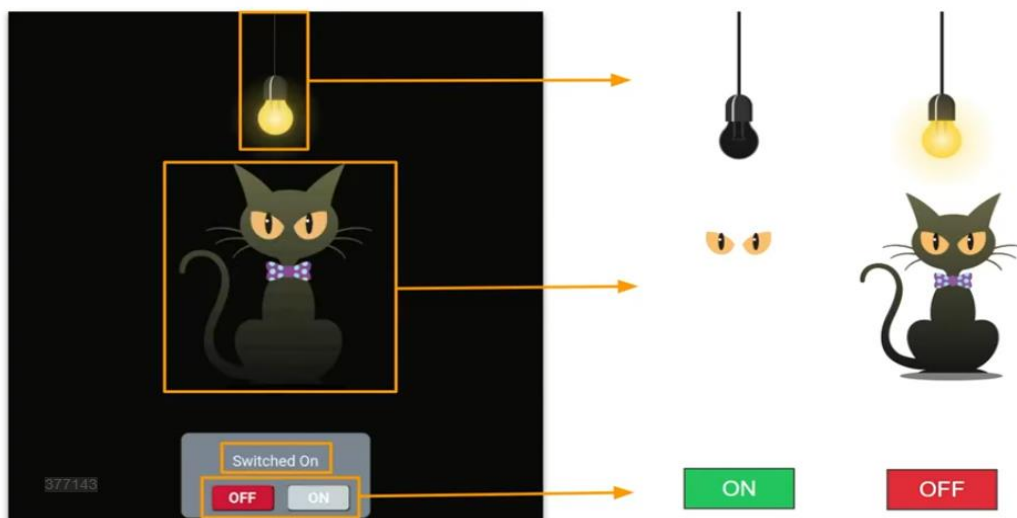


Cat & Light Example



Cat & Light Example

Breaking Down into Parts





How do we manipulate HTML and CSS Using JS?

JavaScript



Document Object Model (DOM)

DOM

Document Object Model

DOM is the **structured representation** of the content in the document created by the browser



DOM

HTML DOM

Tree structured representation of HTML Document created by the browser.

It allows JavaScript to change the document **structure, style** and **content**.



Example

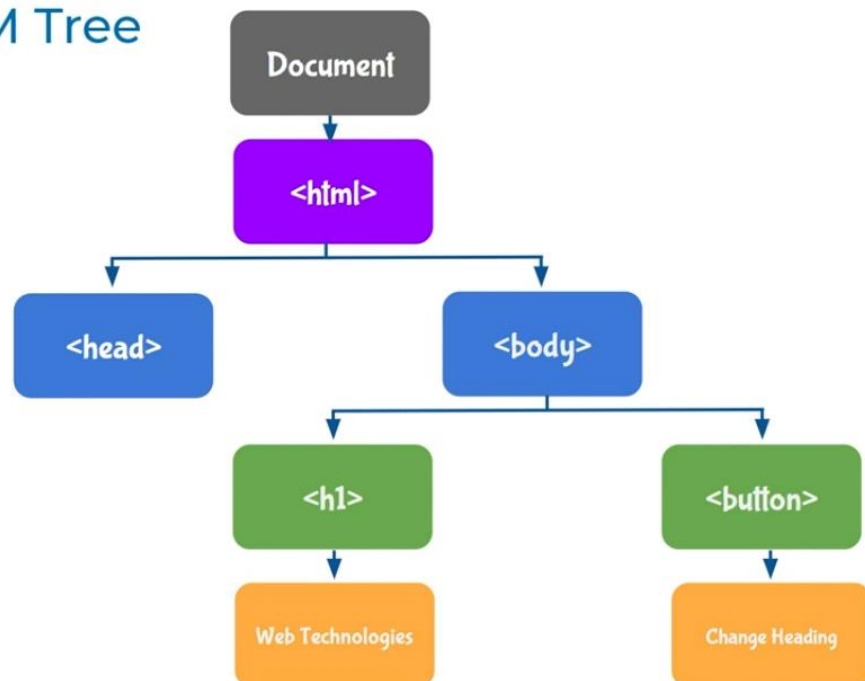
```
<!DOCTYPE html>
<html>
  <head></head>
  <body>
    <h1>Web Technologies</h1>
    <button>Change Heading</button>
  </body>
</html>
```

HTML

Web Technologies

Example

HTML DOM Tree

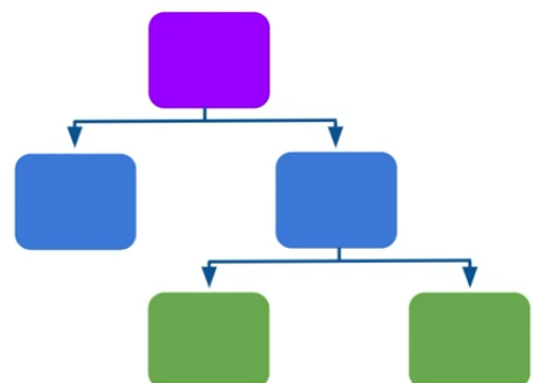


HTML DOM Tree

Nodes

The DOM tree represents
HTML document as nodes.

Each node is referred to as an **Object**



Document Object

It is the **entry point** of the DOM

To access any HTML Element,
you should always start with accessing
the **document** object **first**.

Document

Let's manipulate HTML and CSS
using JavaScript

Web Technologies

Change Heading



4.0 Technologies

Change Heading

Manipulating HTML & CSS

Steps:

- Select the Heading Element
- Manipulate the text content of the Heading Element
- Manipulate the Styles of the Heading Element



Step 1

Selecting the Heading Element

HTML

```
<body>
  <h1 id="headingElement">Web Technologies</h1>
  <button>Change Heading</button>
</body>
```

Step 1

Selecting the Heading Element

JS

```
document.getElementById("headingElement")
```

The `getElementById()` method helps to select the HTML Element with a **specific Id**

Step 1

Selecting the Heading Element

JS

```
console.log(document.getElementById("headingElement"));
```

Step 1

Selecting the Heading Element

JS

```
console.log(document.getElementById("headingElement"));
```

Output

```
<h1 id="headingElement">Web Technologies</h1>
```


Step 2

Manipulating the Text Content

JS

```
document.getElementById("headingElement").textContent = "4.0 Technologies";
```

To manipulate the text within the HTML Element
use `textContent` Property

Manipulating the Text Content

Output

Web Technologies

Change Heading



4.0 Technologies

Change Heading

Step 3

Manipulating the Styles

JS

```
document.getElementById("headingElement").style.color = "blue";
```

The `style` property is used to get or set a **specific style**
of an HTML Element using different **CSS properties**

Output

4.0 Technologies

Change Heading



4.0 Technologies

Change Heading

How to manipulate
HTML and CSS

Based on User Actions?

4.0 Technologies

Change Heading



Events

DOM

Events

Events are the actions by which the User or Browser **interact** with HTML Elements

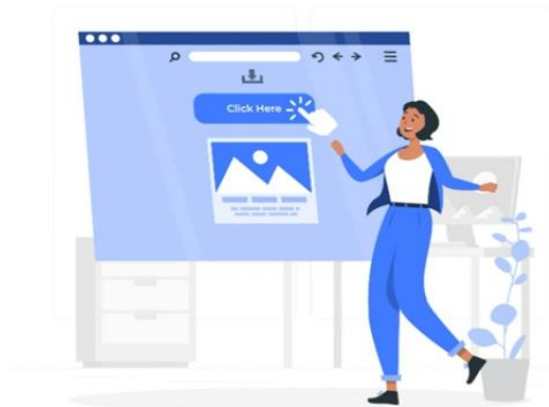
Actions can be:

- **clicking** button,
- **pressing** keyboard keys,
- **scrolling** the page, etc



onclick Event

onclick events occur when user **clicks** on an HTML Element



HTML

The onclick Attribute

HTML

```
<body>
  <h1 id="headingElement">Web Technologies</h1>
  <button onclick="manipulateStyles()">Change Heading</button>
</body>
```

onclick Event

Defining a Function

**def** function_name():

Reusable Code Block

**function** functionName() {

Reusable Code Block

}

onclick Event

Defining a Function

JS

```
function manipulateStyles() {
  console.log("Hi, this is Rahul");
}
```

Output

Hi, this is Rahul

Manipulating Styles

JS

```
function manipulateStyles() {  
  document.getElementById("headingElement").textContent = "4.0 Technologies";  
  document.getElementById("headingElement").style.color = "blue";  
}
```

Output

Web Technologies

Change Heading



4.0 Technologies

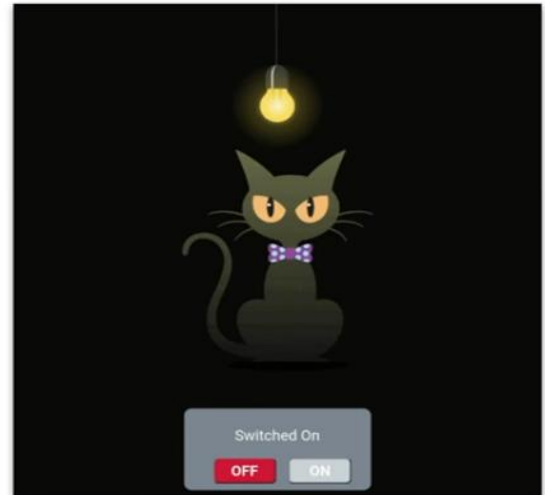
Change Heading

Let's build the
Cat & Light
Example



Observations

- Switch Status changes
- Bulb goes On and Off
- Cat become Visible and Invisible
- Switch (button) background color Changes



HTML & CSS Code

```
<div class="dark-background text-center">
  ...
  <div class="d-flex flex-row justify-content-center pt-5">
    <div class="switch-board">
      ...
    </div>
  </div>
</div>
```

HTML

HTML Code:

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
  <meta charset="UTF-8">
```

```
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
  <title>Document</title>
```

```
  <link rel="stylesheet" href="cat.css"/>
```

```

<script src="./cat.js"></script>

</head>

<body>

  <div class="dark-background text-center">

    <div>

    </div>

    <div>

    </div>

    <div class="d-flex flex-row justify-content-center pt-5">

      <div class="switch-board">

        <h1 class="switch-status" id="switchStatus">Switched On</h1>

        <button class="off-switch" id="off-Switch" onclick="switchOff()">
OFF</button>

        <button class="on-switch" id="onSwitch" onclick="switchOn()">ON
</button>

      </div>

    </div>

  </div>

</body>

</html>

```

CSS:

```

@import
url("https://fonts.googleapis.com/css2?family=Bree+Serif&family=Caveat:wght@4
00;700&family=Lobster&family=Monoton&family=Open+Sans:ital,wght@0,400;0,
700;1,400;1,700&family=Playfair+Display+SC:ital,wght@0,400;0,700;1,700&family

```

=Playfair+Display:ital,wght@0,400;0,700;1,700&family=Roboto:ital,wght@0,400;0,700;1,400;1,700&family=Source+Sans+Pro:ital,wght@0,400;0,700;1,700&family=Work+Sans:ital,wght@0,400;0,700;1,700&display=swap");

.dark-background{

background-color: #0b0b0b;

}

.bulb-image

{

width: 150px;

margin-left: 90px;

}

.cat-image

{

width: 300px;

}

.switch-board{

background-color: #7b8794;

width: 294px;

height: 139px;

border-radius: 20px;

padding-left: 16px;

padding-right: 16px;

padding-top: 16px;

padding-bottom: 32px;

margin: 16px;

}

.switch-status

```
{  
  color: #ffffff;  
  font-family: "Roboto";  
  font-size: 30px;  
  font-weight: 500;  
  margin-bottom: 24px;  
  text-align: center;  
  
}
```

.on-switch

```
{  
  color: #ffffff;  
  background-color: green;  
  font-family: "Roboto";  
  font-size: 24px;  
  font-weight: bold;  
  width: 99px;  
  height: 44px;  
  border-radius: 8px;  
  margin-left: 32px;  
  
}
```

.off-switch

```
{  
  color: #ffffff;  
  background-color: #e12d39;  
  font-family: Roboto;  
  font-size: 24px;  
  
}
```



```
font-weight: bold;  
width: 99px;  
height: 44px;  
border-radius: 8px;  
margin-left: 32px;  
}
```

How to achieve this?



DOM and Event Fundamentals

Cat & Light Example

Steps:

- Make Switches listen to User Actions
- Make bulb Go On and Off
- Make cat Visible and Invisible
- Update Switch Status
- Change the background color of the switch



Step 1

Making Switches listen to User Actions

```
...
<div class="switch-board">
  <h1 class="switch-status">Switched on</h1>
  <button class="off-switch" onclick="switchOff()">
    OFF
  </button>
  <button class="on-switch" onclick="switchOn()">
    ON
  </button>
</div>
...
```

Step 1

Making Switches listen to User Actions

```
function switchOff() {

}

function switchOn() {

}
```

How to make Bulb Go On and Off?



Step 2

Making Bulb Go On and Off



Switch On



Switch Off

Making Bulb Go On and Off

Specifying id

```
<div class="dark-background text-center">
  <div>
    
  </div>
  ...
</div>
```

HTML

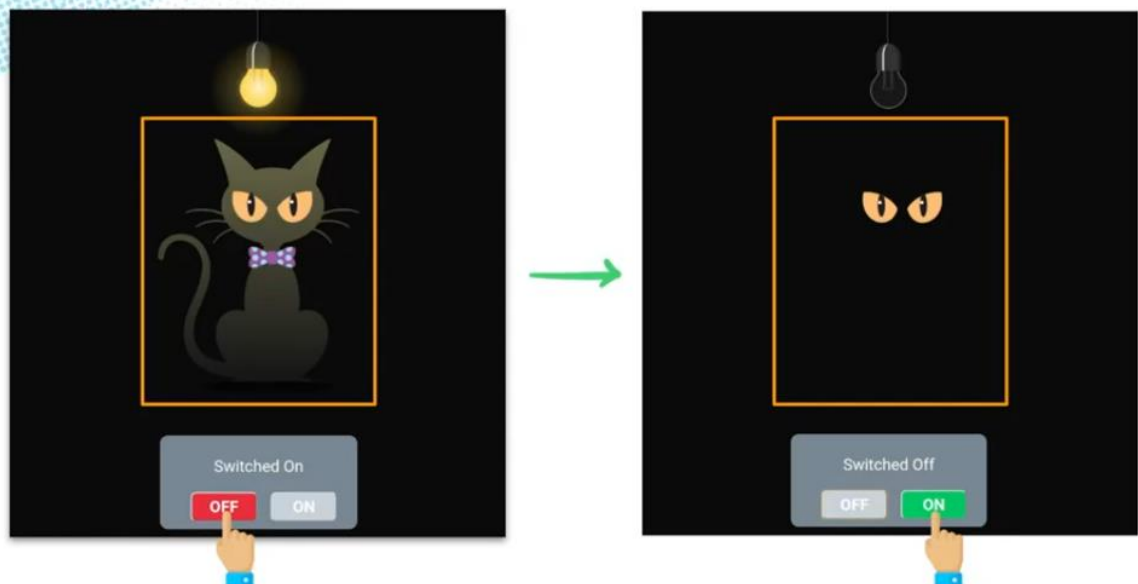
Making Bulb Go On and Off

Changing the Bulb Image Sources

```
function switchOff() {
  document.getElementById("bulbImage").src = "IMAGE_URL";
}
function switchOn() {
  document.getElementById("bulbImage").src = "IMAGE_URL";
}
```

JS

Let's make Cat Visible and Invisible



Step 3

Making Cat Visible and Invisible



Switch On



Switch Off

Making Cat Visible and Invisible

Specifying id

```
<div class="dark-background text-center">
  ...
  <div>
    
  </div>
  ...
</div>
```

HTML

Change the Cat Image Sources

JS

```
function switchOff() {
  ...
  document.getElementById("catImage").src = "IMAGE_URL";
}
function switchOn() {
  ...
  document.getElementById("catImage").src = "IMAGE_URL";
}
```

Step 4 - Changing the Switch Status

Specifying id

HTML

```
<div class="dark-background text-center">
  ...
  <div class="d-flex flex-row justify-content-center pt-5">
    <div class="switch-board">
      <h1 class="switch-status" id="switchStatus">Switched on</h1>
      ...
    </div>
  </div>
</div>
```

Changing the Switch Status

Changing the Text Content

JS

```
function switchOff() {
  ...
  document.getElementById("switchStatus").textContent = "Switched Off";
}
function switchOn() {
  ...
  document.getElementById("switchStatus").textContent = "Switched On";
}
```

JS:

function switchOff()

{

document.getElementById("bulbimage").src="https://d1tgh8fmlzexmh.cloudfront.net/ccbp-dynamic-webapps/bulb-go-off-img.png";

document.getElementById("catimage").src="https://d1tgh8fmlzexmh.cloudfront.net/ccbp-dynamic-webapps/cat-eyes-img.png";

document.getElementById("switchStatus").textContent="SwitchedOff";

document.getElementById("onSwitch").style.backgroundColor="#22c55e";

document.getElementById("offSwitch").style.backgroundColor="#cbd2d9";

}

function switchOn()

{

document.getElementById("bulbimage").src="https://d1tgh8fmlzexmh.cloudfront.net/ccbp-dynamic-webapps/bulb-go-on-img.png";

document.getElementById("catimage").src="https://d1tgh8fmlzexmh.cloudfront.net/ccbp-dynamic-webapps/cat-img.png";

document.getElementById("switchStatus").textContent="SwitchedOn"

}